

Santosh S Data Engineer

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Summary

Data Engineer with 5+ years of experience in designing, building, and optimizing data platforms and pipelines across cloud and on-premise environments. Proficient in real-time and batch processing, ETL automation, data warehousing, and data governance. Skilled in Python, SQL, Spark, Kafka, Airflow, and cloud platforms (AWS, Azure, GCP, Snowflake) to deliver secure, scalable, and high-performance data solutions.

Technical Skills

- Programming & Scripting:** Python (pandas, pyjanitor, Polars), SQL (window functions, CTEs, indexing, partitioning, caching), R, Bash, Shell, PySpark
- Data Architecture & Modeling:** Medallion Architecture (Bronze, Silver, Gold), Dimensional Modeling, Star & Snowflake Schemas, Data Vault, Event-Driven Architecture, Microservices, Real-Time & Batch Processing, Data Warehousing
- Big Data & Distributed Systems:** Apache Spark (Catalyst Optimizer, Tungsten Engine, Adaptive Query Execution), Apache Flink, Apache Beam, Dask, Hadoop Ecosystem (HDFS, MapReduce, YARN, Hive, HBase), Presto, Apache Impala, Apache Druid, Spark Streaming
- Data Ingestion & Streaming:** Apache Kafka, Apache NiFi, Apache Pulsar, AWS Kinesis, Flume, Logstash, Sqoop, Spark Structured Streaming
- Cloud Platforms & Services:** AWS (Lambda, Glue, Redshift, EMR, S3, CloudWatch, IAM), Azure (Data Factory, Synapse Analytics, Databricks), GCP (BigQuery, Dataflow, Pub/Sub, Dataproc), Snowflake
- ETL & Workflow Orchestration:** Apache Airflow, Prefect, dbt, Talend, Luigi, Databricks ETL, CI/CD Pipelines
- Data Governance & Quality:** Great Expectations, Monte Carlo, Apache Atlas, OpenLineage, Automated Data Lineage, Metadata Management, GDPR, HIPAA, CCPA, Data Validation & Monitoring
- NoSQL & Search Databases:** MongoDB, Cassandra, Elasticsearch, Redis, DynamoDB
- Machine Learning & AI:** PySpark MLlib, TensorFlow, Scikit-learn, Hugging Face, Feature Engineering, NLP Datasets, MLflow, Gen AI
- Data Security & Compliance:** Encryption, RBAC, Data Masking, SOC 2, PCI DSS, Network Security, Secure Cloud Integrations

Professional Experience

Data Engineer, Visa

10/2024 – Present | Remote, USA

- Worked on the AI-Enabled Global Payments Analytics & Data Pipeline Platform, collaborating with Product, IT, Risk, and Analytics teams, gathering requirements, creating documentation, and executing secure, regulatory-aligned workflows, improving client satisfaction metrics by 35%.
- Built real-time payment transaction and fraud detection pipelines using Kafka Streams, AWS Lambda, and Spark Streaming, using automated anomaly detection for suspicious transactions and compliance alerts, reducing issue resolution time ensuring end-to-end encryption and secure data access.
- Developed and automated ETL workflows using Apache Airflow, Prefect, Pandas, and Python, applying window functions, CTEs, dynamic partitioning, and hybrid batch+streaming pipelines for customer segmentation and payment analytics, increasing pipeline efficiency and trust in data by 30%.
- Created scalable analytics layers using Amazon Redshift, Athena, and ClickHouse, delivering sub-second dashboards and real-time KPI reporting on payment volumes, transaction risk, and loyalty metrics, maintaining 99.9% uptime, ensuring compliance, and providing secure insights to build enterprise-level trust in transaction data.
- Prepared high-quality training datasets for AI-powered fraud detection and consumer behavior analysis, using Snowflake, AWS Glue, and PySpark, securely processing payment logs and user interactions, improving model accuracy by 50%, reducing preprocessing time.
- Ensured end-to-end observability and anomaly detection in data workflows, collaborating with Engineering, Data Ops, and Compliance teams to maintain secure AWS integration and build trustworthy, audit-ready analytics pipelines.

Data Engineer, Capgemini

12/2020 - 08/2023 | Hyderabad, India

- Architected and implemented a cloud-based AWS data platform to centralize patient and claims data, ensuring analytics, reporting, and compliance requirements were met efficiently, providing stakeholders with unified and secure access to critical healthcare information.
- Developed scalable ETL pipelines using PySpark and AWS Glue, ingesting and transforming over 5 TB of daily claims data, and loading curated datasets into Amazon Redshift, enabling consistent analysis for care management, finance, and compliance teams.
- Led migration of legacy SSIS-based data warehouse to AWS Glue and Redshift, reducing data-refresh time by 45% and cutting infrastructure costs by \$6,000 annually, while maintaining HIPAA compliance and improving overall data accessibility and reliability.
- Optimized data processing by 40% with Spark on AWS EMR, parallelizing claims processing and automating validation workflows, ensuring high-quality, accurate, and timely healthcare metrics for reporting, operational efficiency, and decision-making purposes.
- Delivered machine learning-ready datasets to identify recurring claims rejection patterns, enabling proactive denial reduction, improving reimbursement cycles, and helping business teams decrease denial rates by 15%, driving actionable insights and operational efficiency.

Data Engineer, Sutherland Global

05/2019 - 12/2020 | Remote, India

- Built end-to-end fraud detection pipelines using Azure Data Factory, Synapse, and Data Lake to process 5M+ daily e-commerce transactions, securing \$100M+ revenue and achieving 99% transaction accuracy across critical business operations.
- Designed and optimized Azure Synapse data marts and schemas, automating order reconciliation and data validation with Python and serverless Azure Functions, cutting query times by 75% and reducing manual QA by 85% (~15+ hours/week).
- Developed real-time Power BI dashboards via Azure DevOps, enabling leadership to monitor fraud detection, sales, and supply chain events, reducing detection time by 60% and providing actionable insights for rapid operational decision-making.

Education

University of New Haven, New Haven, CT, USA

Master of Science in Data Science

08/2023 – 05/2025

Dr. B.R. Ambedkar University, Srikakulam, India

Bachelor of Science in Computer Science

06/2017 – 05/2020

Certifications

AWS Certified Data Engineer – Associate | [eb63ce56ed5b46a2bd38b9ef1e517d57](#)

Leadership & Volunteering Experience

- Led a 6-member team for the Capstone Project at University of New Haven, delivering forecasting models for enrollment trends.
- Served as Program Coordinator for the Entrepreneur Cell and actively contributed as a Student Council member during undergrad.
- Led a social media campaign to promote the college cultural festival, boosting event visibility and student engagement.
- Active member of Toastmasters International, enhancing public speaking and leadership skills.